

$$\int_{-1}^1 \frac{f(t)}{\sqrt{1-t^2}} dt = A_0 f(-1) + A_1 f(t_1) + A_2 f(t_2) + A_3 f(t_3) + A_4 f(1) + R f$$

← fixat first

← Gauss-Lobatto

• Gauss-Lobatto

$$\int_a^b f(x) w(x) dx - A_1 \cdot f(a) - \sum_{i=2}^{n-1} A_i \cdot f(x_i) - A_n \cdot f(b) = \frac{f^{(2n-2)}(\xi)}{(2n-2)!} \int_a^b \pi_{n-2}^2(x) \cdot (x-a) \cdot (b-x) \cdot w(x) dx$$

← $q_1(x) = 2n-2$

π_{n-2} = polinomul ortogonal (în raport cu ponderea $w_{a,b}(x) = (x-a)(b-x)w(x)$) de grad $n-2$ care este monic (coeficientul lui x^{n-2} este 1, i.e. $\pi_{n-2}(x) = (x-x_2) \cdot \dots \cdot (x-x_{n-1})$).

$$\pi_3 = \text{pol. ort. în rap. } w_{a,b}(x) = w_{-1,1}(x) = (1+x)(1-x) w(x) = (1-x^2) \cdot \frac{1}{\sqrt{1-x^2}} = \sqrt{1-x^2}$$

↓

Gauss-Lobatto II

• Relația de recurență este

$$Q_{n+1}(t) = 2tQ_n(t) - Q_{n-1}(t), \quad Q_0(t) = 1, \quad Q_1(t) = 2t.$$

← log. pdf

$$\hookrightarrow \text{para } (w(-x) = w_{a,b}(x))$$

$(1-t^2)^{1/2}$	$[-1, 1]$	Chebyshev #2	α_n	β_0	$\beta_n, n > 0$
			0	$\frac{1}{2}\pi$	$\frac{1}{4}$

$$\bar{u}_{n+1}(t) = (t - \alpha_n) \bar{u}_n(t) - \beta_n \cdot \bar{u}_{n-1}(t), \quad \bar{u}_{-1} = 0, \quad \bar{u}_0 = 1$$

$$\bar{u}_{n+1}(t) = t \bar{u}_n(t) - \frac{1}{n} \bar{u}_{n-1}(t)$$

$$\bar{u}_1 = t \cdot \bar{u}_0 = t$$

$$\bar{u}_2 = t \cdot \bar{u}_1 - \frac{1}{1} \bar{u}_0 = t^2 - \frac{1}{4}$$

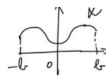
$$\bar{u}_3 = t \cdot \bar{u}_2 - \frac{1}{2} \bar{u}_1 = t^3 - t \cdot \frac{1}{4} - \frac{1}{2} t = t^3 - \frac{1}{2} t$$

$$\bar{u}_3(t) = 0 \Leftrightarrow t^3 - \frac{1}{2} t = 0 \Leftrightarrow t(t^2 - \frac{1}{2}) = 0 \Leftrightarrow \{t_1, t_2, t_3\} = \{0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\}$$

$$t_1 = -\frac{1}{\sqrt{2}}, \quad t_2 = 0, \quad t_3 = \frac{1}{\sqrt{2}}$$

Obs.: $a = -b$ și $w(-x) = w(x), x \in (-b, b)$

$$\Rightarrow \begin{cases} \alpha_n = 0 \\ \beta_n(-x) = (-1)^n \beta_n(x) \end{cases}, \quad n = 0, 1, 2, \dots$$



$$\Rightarrow A_1 = A_3$$

$$\Rightarrow \begin{cases} L_{n-1/2} = -L_n \\ A_{n-1/2} = A_n \end{cases}, \quad L_2 = \pi, u.$$

$$\Rightarrow A_0 = A_4$$

$$f(t) \equiv 1 \Rightarrow \int_{-1}^1 \frac{1}{\sqrt{1-t^2}} dt = A_0 \underbrace{f(-1)}_1 + A_1 \underbrace{f(-\frac{1}{\sqrt{2}})}_1 + A_2 \underbrace{f(0)}_1 + A_1 \underbrace{f(\frac{1}{\sqrt{2}})}_1 + A_0 \underbrace{f(1)}_1$$

$$\arcsin x \Big|_{-1}^1 = \pi$$

$$2A_0 + 2A_1 + A_2 = \pi$$

$$f(t) = t \Rightarrow \int_{-1}^1 \frac{t}{\sqrt{1-t^2}} dt \stackrel{\substack{\text{fkt. symmetrisch} \\ \text{interval symmetrisch}}}{=} 0 = A_0(-1) + A_1(-\frac{1}{\sqrt{2}}) + A_2 \cdot 0 + A_1(\frac{1}{\sqrt{2}}) + A_0 \cdot 1$$

$$f(t) = t^2 \Rightarrow \int_{-1}^1 \frac{t^2+1}{\sqrt{1-t^2}} dt = \int_{-1}^1 \frac{t}{\sqrt{1-t^2}} \cdot t dt + \int_{-1}^1 \frac{1}{\sqrt{1-t^2}} dt = \underbrace{\left(\sqrt{1-t^2} \right) \cdot t \Big|_{-1}^1}_{=0} + \int_{-1}^1 \frac{1}{\sqrt{1-t^2}} dt$$

$$\left(\sqrt{1-t^2} \right)' = -\frac{1}{2} \cdot \frac{1}{\sqrt{1-t^2}} \cdot (-2t)$$

$$\int_{-1}^1 \frac{1}{\sqrt{1-t^2}} dt = \int_{-1}^1 \sqrt{1-t^2} dt = \pi - \frac{\pi}{2} = \frac{\pi}{2} = \boxed{\frac{\pi}{2}}$$

$$\begin{aligned} & \xrightarrow[t \in [0, \pi]]{t = \cos y} \int_{\pi}^0 \sqrt{1-\cos^2 y} \cdot (-\sin y) dy \\ &= \int_0^{\pi} \sin^2 y dy = \int_0^{\pi} \frac{1}{2}(1 - \cos(2y)) dy \\ &= \frac{\pi}{2} - \frac{\sin(2y)}{2} \Big|_0^{\pi} = \frac{\pi}{2} \end{aligned}$$

$$\frac{\pi}{2} = A_0 \cdot 1 + A_1 \cdot \frac{1}{2} + A_1 \cdot \frac{1}{2} + A_0 \cdot 1$$

$$2A_0 + A_1 = \frac{\pi}{2}$$

$$\begin{cases} 2A_0 + A_1 = \frac{\pi}{2} \\ 2A_0 + 2A_1 + A_2 = \pi \end{cases}$$

$$f(t) = t^4 \Rightarrow \int_{-1}^1 \frac{t^4}{\sqrt{1-t^2}} dt \stackrel{t = \cos y}{=} \int_{\pi}^0 \frac{\cos^4 y}{\sin y} \cdot (-\sin y) dy = \int_0^{\pi} \cos^4 y dy$$

$$= \int_0^{\pi} \left(\frac{1}{2}(1 + \cos(2y)) \right)^2 dy = \frac{1}{4} \int_0^{\pi} (1 + 2\cos(2y) + \cos^2(2y)) dy$$

$$= \frac{1}{4} \left(\pi + \sin(2y) \Big|_0^{\pi} + \frac{\pi}{2} + \frac{\sin(4y)}{4} \Big|_0^{\pi} \right) = \frac{3\pi}{8} = A_0 + A_1 \cdot \frac{1}{4} + A_1 \cdot \frac{1}{4} + A_0$$

$$\int_0^{\pi} \cos^n y dy = \dots$$

$$\Rightarrow 2A_0 + \frac{1}{2} A_1 = \frac{3\pi}{8}$$

$$\frac{1}{2} A_1 = \frac{\pi}{2} - \frac{3\pi}{8} = \frac{\pi}{8} \Rightarrow \boxed{A_1 = \frac{\pi}{4}} = A_3$$

$$\frac{1}{2} A_1 = \frac{\pi}{2} - \frac{3\pi}{8} = \frac{\pi}{8} \Rightarrow \boxed{A_1 = \frac{\pi}{4}} = A_3$$

$$2A_0 + \frac{\pi}{8} = \frac{3\pi}{8} \Rightarrow \boxed{A_0 = \frac{\pi}{8}} = A_4$$

$$\frac{\pi}{4} + \frac{\pi}{2} + A_2 = \pi \Rightarrow \boxed{A_2 = \frac{\pi}{4}}$$

$$\int_{-1}^1 \frac{f(t)}{\sqrt{1-t^2}} dt = \frac{\pi}{8} f(-1) + \frac{\pi}{4} f(-\frac{1}{\sqrt{2}}) + \frac{\pi}{4} f(0) + \frac{\pi}{4} f(\frac{1}{\sqrt{2}}) + \frac{\pi}{8} f(1) + Rf$$

II

$$A_k = \int_a^b \frac{(x-x_1) \dots (x-x_{k-1})(x-x_{k+1}) \dots (x-x_n)}{(x_k-x_1) \dots (x_k-x_{k-1})(x_k-x_{k+1}) \dots (x_k-x_n)} \cdot w(x) dx$$

$$A_0 = \int_{-1}^1 \frac{(x+\frac{1}{\sqrt{2}})(x-\frac{1}{\sqrt{2}})(x-1)}{(-1+\frac{1}{\sqrt{2}})(-1)(-1-\frac{1}{\sqrt{2}})(-1-1)} \cdot \frac{1}{\sqrt{1-x^2}} dx \quad t_0=-1, t_1=-\frac{1}{\sqrt{2}}, t_2=0, t_3=\frac{1}{\sqrt{2}}, t_4=1$$

$$\int_{-1}^1 \frac{f(t)}{\sqrt{1-t^2}} dt = A_1 f(0) + B_1 f'(0) + A_2 f(t_2) + R(f) \quad \text{cu } q \cdot \text{ exactitate}$$

$$f=1 \Rightarrow \pi = \int_{-1}^1 \frac{1}{\sqrt{1-t^2}} dt = A_1 + 0 + A_2 \cdot 1$$

$$f(t)=t \Rightarrow 0 = \int_{-1}^1 \frac{t}{\sqrt{1-t^2}} dt = B_1 + A_2 \cdot t_2$$

$$f(t)=t^2 \Rightarrow \dots$$

$$\int_{-1}^1 \frac{f(t)}{\sqrt{1-t^2}} dt = \sum A_i f(t_i) + R(f)$$

$$f = \text{pol. fundamental Lagrange} = u_i(t_j) = \begin{cases} 1, & t_j = t_i \\ 0, & t_i \neq t_j \end{cases}$$

$$\int_{-1}^1 \frac{u_{i0}(t)}{\sqrt{1-t^2}} dt = \sum A_i u_{i0}(t_i) = A_{i0}$$

$$R(f) = ?$$

• Gauss-Lobatto:

$$\int_a^b f(x)w(x)dx = A_1 \cdot f(a) + \sum_{i=2}^{n-1} A_i \cdot f(x_i) + A_n \cdot f(b) = \frac{f^{(2n-2)}(\xi)}{(2n-2)!} \int_a^b \pi_{n-2}^2(x) \cdot (x-a) \cdot (b-x) \cdot w(x) dx$$

π_{n-2} = polinomul ortogonal (în raport cu ponderea $w_{a,b}(x) = (x-a)(b-x)w(x)$) de grad $n-2$ $w_{a,b}(x)$
care este monic (coeficientul lui x^{n-2} este 1, i.e. $\pi_{n-2}(x) = (x-x_2) \cdot \dots \cdot (x-x_{n-1})$).

$$R(f) = \frac{f^{(8)}(\xi)}{8!} \int_{-1}^1 \pi_3^2(x) \cdot \sqrt{1-x^2} dx = \frac{f^{(8)}(\xi)}{8!} \int_{-1}^1 (x^6 - x^4 + \frac{x^2}{4}) \sqrt{1-x^2} dx$$

$$R(f) = \frac{f^{(8)}(\xi)}{8!} \int_{-1}^1 \frac{\pi(\xi) \cdot \sqrt{1-x^2} dx}{\underbrace{\left(x(x^2 - \frac{1}{2})\right)^2}_{x^2(x^4 - x^2 + \frac{1}{4})}} = \frac{f^{(8)}(\xi)}{8!} \int_{-1}^1 (x^6 - x^4 + \frac{x^2}{4}) \sqrt{1-x^2} dx$$

$$\int_{-1}^1 x^6 \sqrt{1-x^2} dx = \dots$$

$$\int_{-1}^1 x^4 \sqrt{1-x^2} dx = \dots$$

$$\int_{-1}^1 x^2 \sqrt{1-x^2} dx = \dots$$

Problema:

Fie α un zero simplu al lui f și $f \in C^p$ în vecinătatea lui α , unde $p \geq 3$.

(a) Arătați că dacă $f^{(k)}(\alpha) = 0$ pentru $k = 2, \dots, p-1$ și $f^{(p)}(\alpha) \neq 0$, atunci metoda lui Newton aplicată lui $f(x) = 0$ converge local către α cu ordinul p . Determinați constanta de eroare asimptotică.

(b) Deduceți o metodă pentru calculul lui $\sqrt[p]{a}$, aplicând rezultatul precedent funcției $f(x) = x^4 - a$. Cât este ordinul de convergență?

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

↳ met. lui Newton
($f'(\alpha) \neq 0$)

$$\text{I } \varphi(x) = x - \frac{f(x)}{f'(x)}$$

$$\text{II } e_n = x_n - \alpha$$

$$x_{n+1} - \alpha = x_n - \alpha - \frac{f(x_n)}{f'(x_n)}$$

$$e_{n+1} = \frac{(x_n - \alpha) f'(x_n) - f(x_n)}{f'(x_n)}$$

$$\approx \frac{(x_n - \alpha)^p \frac{f^{(p)}(\alpha)}{p!} \left(\frac{1}{(p-1)!} - \frac{1}{p!}\right)}{f'(\alpha)}$$

$$\left| \frac{e_{n+1}}{e_n^p} \right| \approx \frac{\left| \frac{f^{(p)}(\alpha)}{p!} \right| \frac{p-1}{p!}}{|f'(\alpha)|}$$

$$\frac{f^{(p)}(\alpha) \cdot (p-1)}{f'(\alpha) \cdot p!} \neq 0 \Rightarrow \text{er. asimp.} \Rightarrow \boxed{\text{ord. } p}$$

$$f(\alpha) = 0, \quad h = 2, p-1$$

$$f^{(p)}(\alpha) \neq 0$$

Definiția 2

Se spune că x_n converge către α (cel puțin) cu ordinul $p \geq 1$ dacă (3) are loc cu

$$\lim_{n \rightarrow \infty} \frac{e_{n+1}}{e_n^p} = c, \quad c > 0 \quad (5)$$

Taylor

$$f(x) \approx \underbrace{f(\alpha)}_{=0} + \frac{f'(\alpha)}{1!} (x-\alpha) + \sum_{k=2}^{p-1} \frac{f^{(k)}(\alpha)}{k!} (x-\alpha)^k$$

$$+ \frac{f^{(p)}(\alpha)}{p!} (x-\alpha)^p$$

$$f(x) \approx f'(\alpha)(x-\alpha) + \frac{f^{(p)}(\alpha)}{p!} (x-\alpha)^p$$

$$f'(x) \approx f'(\alpha) + \frac{f^{(p)}(\alpha)}{(p-1)!} \cdot (x-\alpha)^{p-1} \quad | \cdot (x-\alpha) |$$

$$(x-\alpha) f'(x) - f(x) \approx f^{(p)}(\alpha) (x-\alpha)^p \left(\frac{1}{(p-1)!} - \frac{1}{p!} \right)$$

(b) Deduceți o metodă pentru calculul lui $\sqrt[4]{a}$, aplicând rezultatul precedent funcției $f(x) = x^{4-\lambda} - ax^{-\lambda}$, pentru un λ convenabil. (1p)
Cât este ordinul de convergență? (2p)

$$a > 0$$

$$f(\sqrt[4]{a}) = (\sqrt[4]{a})^{4-\lambda} - a(\sqrt[4]{a})^{-\lambda} = \underbrace{(\sqrt[4]{a})^4}_a \cdot (\sqrt[4]{a})^{-\lambda} - a(\sqrt[4]{a})^{-\lambda} = 0$$

$$f'(x) = (4-\lambda)x^{3-\lambda} + a\lambda x^{-\lambda-1}; \quad f'(\sqrt[4]{a}) = (4-\lambda)a^{\frac{3-\lambda}{4}} + \lambda a^{-\frac{\lambda}{4}} \neq 0 \Leftrightarrow (4-\lambda)a^{\frac{3}{4}} + \lambda a^{\frac{3}{4}} \neq 0$$

$$f''(x) = (4-\lambda)(3-\lambda)x^{2-\lambda} - a\lambda(\lambda+1)x^{-\lambda-2} \quad \Leftrightarrow \lambda(1-a^{\frac{3}{4}}) \neq -4a^{\frac{3}{4}}$$

$$f''(\sqrt[4]{a}) = (4-\lambda)(3-\lambda)a^{\frac{2-\lambda}{4}} - a\lambda(\lambda+1)a^{-\frac{\lambda+2}{4}} = 0 \Leftrightarrow (4-\lambda)(3-\lambda) = \lambda(\lambda+1)$$

$$\Leftrightarrow \lambda^2 - 7\lambda + 12 = \lambda^2 + \lambda$$

$$\Leftrightarrow \lambda = \frac{12}{8} = \frac{3}{2}$$

$$\lambda = \frac{3}{2}$$

$$\downarrow$$

$$f''(x) = \frac{5}{2} \cdot \frac{3}{2} x^{2-\lambda} - a \frac{5}{2} \cdot \frac{3}{2} x^{-\lambda-2} = \frac{15}{4} (x^{\frac{1}{2}} - a x^{-\frac{7}{2}})$$

$$f'''(x) = \frac{15}{4} \left(\frac{1}{2} x^{-\frac{1}{2}} + a \frac{7}{2} x^{-\frac{9}{2}} \right); \quad f'''(\sqrt[4]{a}) = \frac{15}{4} \left(\frac{1}{2} a^{-\frac{1}{8}} + \frac{7}{2} a^{1-\frac{9}{8}} \right)$$

$$= 15a^{-\frac{1}{8}} \neq 0$$

$$\Rightarrow \boxed{p=3} \text{ ord. de conv.}$$